SentiCore

An emotionally intelligent AI that listens, learns, and responds with real empathy

Guide Name
Ms. Philo Sumi
Assistant Professor
Department of AIDS

Team 11 (Batch 1) :
Marc George (MUT22CA043)
Merlin Sarah Jiju (MUT22CA045)
Timon K. John (MUT22CA068)

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Problem Statement

Problem

Traditional emotion-aware AI systems merely detect human affect from text, audio, or vision, but they do not internally simulate or evolve emotional states over time. They lack temporal memory, contextual empathy modeling, and generate only reactive outputs, resulting in one-dimensional, non-adaptive interactions. This limitation prevents such systems from maintaining a coherent emotional "persona" that can learn from past interactions and adjust its responses in future conversations.

Solution

SentiCore, An emotionally intelligent AI that listens, learns, and responds with real empathy

Objective

- The objective of this project is to build an emotionally intelligent AI system that doesn't just recognize emotions ,it actually feels and evolves its emotional state over time, like a human would.
- The system integrates brain-inspired models called Spiking Neural Networks (SNNs), processes speech tone, facial expressions, and text sentiment to understand how a person feels.
- It then uses that understanding to generate emotionally appropriate, empathetic responses — not just one-time replies, but ones that take into account the past conversation and adjust over time.
- To create an emotionally intelligent agent capable of supporting applications such as digital companions, mental health assistants, and adaptive tutors.

Existing System

- **Replica :**
  - **Description:** AI companion chatbot using transformer-based models for conversation.
  - **Key features:** Personality simulation, conversational memory.
  - **Limitation:**No neuromorphic processing, lacks true emotional state evolution.
- **Woebot :**
  - **Description:** Mental health chatbot using cognitive behavioral therapy principles.
  - **Key features:** Mood tracking, therapeutic responses.
  - **Limitation:** Rule-based emotional understanding, no multimodal emotion sensing.
- **Hume AI :**
  - **Description:** Empathic voice interface that analyzes speech prosody for emotion.
  - **Key features:** Real-time voice emotion detection, adaptive responses.
  - **Limitation:**No spiking neural network foundation, limited emotional memory persistence.

Disadvantages of Existing Systems

- All these systems use traditional artificial neural networks and soes not leverage the qualities of SNN
- These system do not take multimodal inputs into consideration and stick on to single model or superficial multimodal inputs.
- Existing systems only react to what you say - they don't internally "feel" or simulate emotions where responses are pattern-matched or statistically generated, not emotionally reasoned.
- High energy consumption per interaction and expensive inference costs limit accessibility and scalability.

Proposed System

- The proposed system converts inputs like tone of voice and the emotional meaning of text into synchronized spike patterns using a neuromorphic-inspired encoding approach and feature extraction.
- A spiking neural core then integrates these spike patterns over time, simulating a continuously evolving emotional state.
- An affective appraisal module compares incoming patterns with learned emotional examples to decide whether to reflect the user's feelings or respond with supportive guidance.
- Finally, the current state of the spiking network, along with the appraisal result, guides a language model to generate responses that are both meaningful and emotionally appropriate.

Feasibility Study (1)

- **Technical Feasibility**
  - SentiCore integrates all core - components (PyTorch, SpikingJelly, Transformers) which are mature and well-documented.
  - Can run on standard GPU infrastructure; neuromorphic hardware (Intel Loihi) available for optimization.
  - Similar multimodal pipelines exist in research (emotion recognition + text generation)
- **Economic Feasibility**
  - Use of open -source libraries reduces development costs
  - SNN deployment can reduce energy costs by 10-100$\times$ vs traditional ANNs.

Feasibility Study (2)

- **Operational Feasibility**
  - The system is highly relevant for target applications such as digital companions, mental health assistants, and adaptive tutors, offering an explainable and adaptive framework.
  - Multimodal Input, Evolving Emotional Memory, Adaptive Empathy, and Emotion-Driven Responses—underscore its operational utility.
  - Each module (audio, text, backend) can be developed in parallel.
- **Legal Ethical Feasibility**
  - Does not create unhealthy dependencies and imposes transparency that Users should know they're interacting with AI.
  - System complies with general data protection practices.
- **Conclusion:**
  - SentiCore is technically implementable, cost-effective, and operationally practical following all legal and ethical practices.

Advantages of the Proposed System

- **Multimodal Input**
  - Includes speech tone and text sentiment into the model for richer emotion understanding.
- **Neuromorphic Emotion Reasoning**
  - Helps the model understand, feel, and reason about human emotions using brain-like spikes and signals that mimic how we recognize emotions.
- **Evolving Emotional Memory**
  - Simulates a continuously changing mood that builds on past interactions for more coherent replies.
- **Adaptive Empathy**
  - Learns when to mirror your feelings or offer comfort by adjusting its behavior based on feedback.
- **Emotion-Driven Responses**
  - Uses its internal mood state to guide the chatbot so responses feel consistent and empathetic.

Software Requirements

- **Python** — Core language for SNN modeling, data processing, and API services.
- **PyTorch** — Deep learning framework powering surrogate-gradient training of spiking networks.
- **SpikingJelly/SNNTorch** — High-performance library for building and training LIF-based SNN layers on GPU.
- **Hugging Face Transformers** — Pretrained GPT-style models for generative, tone-conditioned text output.

Modules

- **Spike Generation Module**
  - Transforms preprocessed text and audio features into spike trains for SNN input.
- **Audio Classification Module**
  - This module is responsible for capturing the user's audio input and transforming it into a format suitable for the spiking neural network (SNN).Classifies audio-derived spike patterns into emotion categories using SNNs and detects emotion states.
- **Chatbot Module**
  - This final module handles the translation of the classification result into a meaningful, actionable response that is delivered back to the user.
- **User Interface Module**
  - Web-based interface (Next.js with typescipt) enabling real-time user interaction with the system.

DFD Level 0

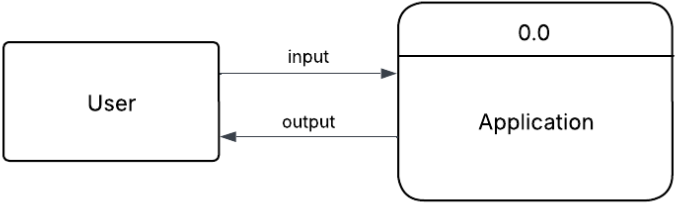

Fig. 1.0: Level 0 Data Flow Diagram

DFD Level 1

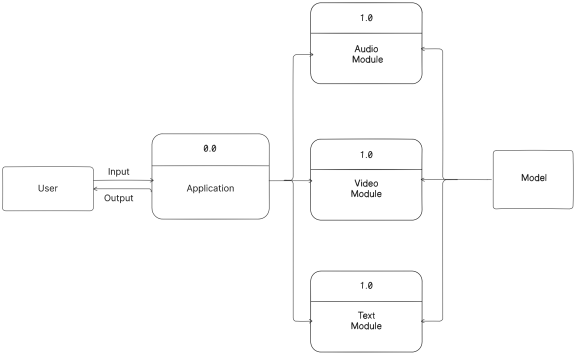

Fig. 1.1: Level 1 Data Flow Diagram

DFD Level 2

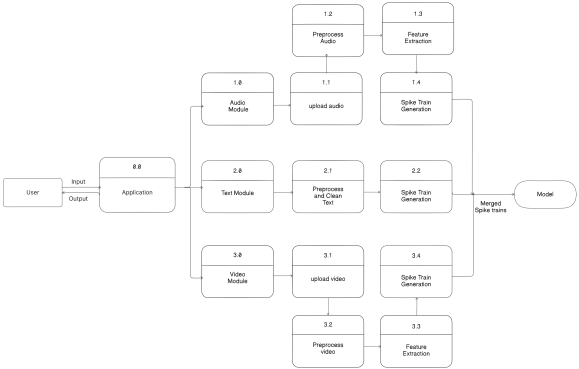

Fig. 1.2: Level 2 Data Flow Diagram

Architecture Diagram

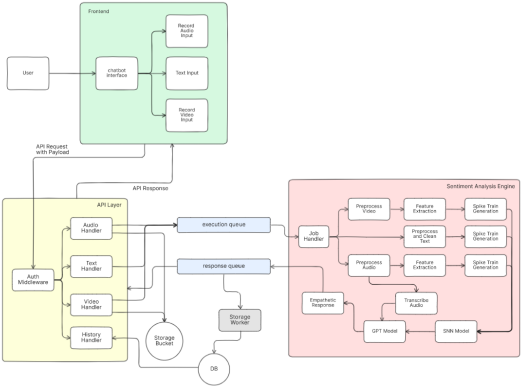

Fig. 2:Architecture diagram

Algorithm/Technique

Technique used is:SNN (Spiking Neueral Network)

Why choose SNN?This method provides better accuracy to the system. Most of the systems use traditional ANN and CNN algorithms,we have levereged spiking neural network which is inspired from the human brain architecture,that fully considers the intrinsic neurophysiological characteristics.

Data Preprocessing Augmentation Pipeline

Data Preprocessing Pipeline

- Load embeddings and labels from dataset files.
- Merge train/val/test dictionaries into a unified feature set.
- Assign unique utterance IDs for each dialogue segment.
- Normalize and clip feature vectors for stable model input
- Stratified split to maintain class balance across Train/Val/Test

Spike Encoding (Rate Coding)

- Convert continuous features into spike trains over multiple timesteps.
- Higher feature values $\rightarrow$ higher firing probability which inturn produces binary spike patterns representing temporal activity.

Data Preprocessing Augmentation Pipeline

Data Augmentation(train only)

- Temporal Jitter: shift spike timings to handle temporal variation.
- Noise Injection: add slight randomness for robustness.
- Temporal Masking: simulate missing or corrupted timesteps.

Batching DataLoader

- Batch spike sequences, labels, and masks in a consistent shape.
- Custom collate function prepares inputs for the SNN.
- Training loader uses weighted sampling; Val/Test use sequential loading.

Audio Processing Inference Pipeline

Audio Input Loading

- User uploads an audio file through FastAPI (multipart request).
- Audio bytes are decoded into a waveform using torchaudio.load()
- Output: waveform tensor and sample rate that supports multiple formats (WAV/MP3/FLAC/OGG, etc.)

Mel-Spectrogram Extraction

- Waveform transformed into a Mel-scale spectrogram.
- Captures human-perceived frequency features
- Output: 2D time–frequency representation (Mel bins $\times$ frames).

Feature Normalization

- Normalize feature vector for stable model behavior and clips values to bounded range.

Audio Processing Inference Pipeline

SNN Inference

- Forward pass through SpikeNet:Embedding → Spiking layers → Attention → Pooling → Classifier
- Audio bytes are decoded into a waveform using torchaudio.load()
- Output: logits for emotion categories

Prediction Label Mapping

- Select highest logit as final class.
- Convert class index to emotion label
- Output: Predicted emotion (e.g., Happy, Neutral, Angry, etc.)

Response Handling

- Exception-safe inference wrapped in try/except
- Format: "response": "Model predicts: <emotion>"

Conclusion

- SentiCore introduces a neuromorphic, spike-based core that includes speech and text into a single emotional stream.
- Its SNN model yields a self-evolving "mood," enabling more coherent, persona-driven dialogue.
- An RL-trained appraisal module dynamically chooses between empathetic mirroring and supportive guidance.
- Conditioning a GPT-style generator on the SNN's state produces responses that are both contextually relevant and emotionally resonant.
- SentiCore offers an explainable, adaptive framework for digital companions, tutors, and mental-health assistants.

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Thank You!